SYED WASAM JAFFRI

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SUMMARY

Final-year AI undergraduate with hands-on experience shipping production AI systems: LLM-backed backend services, RAG pipelines, and ML platforms with measurable cost, latency, and accuracy improvements. Comfortable owning a project end-to-end from data pipeline and model selection through FastAPI deployment and client delivery.

TECHNICAL SKILLS

Programming & Backend: Python, Java, C, SQL, object-oriented design, REST API development (Django, FastAPI, Uvicorn), asynchronous programming

AI & Generative AI: OpenAI API, LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), FAISS & vector databases, HuggingFace, LangChain-equivalent pipelines, PyTorch, Flan-T5, BAAI embeddings

Machine Learning: scikit-learn, XGBoost, LightGBM, model calibration & threshold optimization, feature engineering, ROC-AUC/PR-AUC/lift@k evaluation, SHAP explainability, computer vision (PCA)

Web & Databases: HTML, CSS, JavaScript, React, SQLite, MySQL, data ingestion pipelines

MLOps & Quality: Docker & docker-compose, CI (GitHub Actions), model registry patterns, MLflow/Evidently familiarity, pytest, TDD, pre-commit (black, ruff, isort), mypy, drift monitoring (PSI)

Tools: Git/GitHub, pandas, NumPy, Streamlit

EXPERIENCE

Backend AI Engineer — FlyrankAI, Karachi, Pakistan (remote)

3 months, 2026

Python · FastAPI · OpenAI API · FAISS · HuggingFace · Docker · PostgreSQL

- Cut OpenAI GPT-4 API costs ~30% (monthly token spend) via dynamic context trimming, few-shot compression templates, and token-budget guardrails.
- Boosted retrieval accuracy ~40% over keyword search (precision on held-out queries) by pairing FAISS indexing with OpenAI text-embedding-ada-002.
- Cut inference latency ~35% (P95 response time) by benchmarking 4 LLM configs (GPT-4, GPT-3.5-turbo, Flan-T5, BAAI-bge) and matching models to use-case.
- Shipped 3 production AI microservices, zero downtime, via unified async FastAPI handlers with standardised Pydantic schemas and error handling.
- Lifted response coherence ~25% (internal eval rubric) via structured prompt pipelines with role-based context and chain-of-thought scaffolding.

Freelance Software & AI Developer — Self-employed

2024 – present

- **Tech Stack:** `Java, PHP, JavaScript, JUnit 5, Gson, JSON, REST APIs (QuickChart), Object-Oriented Design (SOLID), CSV Processing, Apache`
- **SpendSmart Finance Tracker:** Delivered a lightweight personal finance web application (PHP, Vanilla JS) featuring role-based authentication, custom expense categorization, and a real-time foreign exchange conversion engine with intelligent caching.
- **Agri-Feed Pricing System:** Built a Java console-based analytical tool for agricultural commodities, processing large-scale CSV datasets to deliver 12-month profit/loss forecasting, percentage change metrics, and automated visual chart generation via the QuickChart API.
- **Surgery Management System:** Engineered a highly modular medical scheduling system using Java and strict SOLID principles, implementing custom Dependency Injection (DI) and dual-persistence data stores (Gson/Java Serialization) to achieve a fully unit-testable architecture.

ACADEMIC PROJECTS (VIEW GITHUB FOR MORE)

RAG Document Cataloger 2025

Python · FastAPI · FAISS · PyTorch · HuggingFace (BAAI/bge, Flan-T5) · SQLite · JWT

- Built a custom LangChain-equivalent RAG pipeline hitting sub-2s query latency across 4 document formats (500-doc corpus benchmark), no orchestration framework.
- Reached ~85% retrieval precision (50-query manual eval) using PyTorch-based BAAI/bge embeddings plus Flan-T5-XL for citation-grounded answers.
- Secured 100% of sessions with zero auth-bypass in pen-testing, via JWT, bcrypt (cost factor 12), and two-step email verification.
- Cut perceived load time ~60% with a responsive single-page frontend: streaming indicators, live document detection, inline PDF viewer.

Customer Churn Prediction Platform 2026

Python · scikit-learn · XGBoost · LightGBM · FastAPI · Streamlit · SHAP · Docker

- Lifted simulated profit ~18% over raw-accuracy baselines (cost-sensitive profit matrix) via 30+ engineered features plus threshold optimization across 4 ensemble models.
- Delivered explainable predictions for 100% of flagged high-risk customers using SHAP attribution and a live what-if simulator on a dedicated FastAPI endpoint.
- Cut model-drift detection time to under 5 minutes with a 6-page Streamlit dashboard: real-time PSI monitoring, KPI cards, automated alerts.
- Generated 15 ROI-ranked retention actions per at-risk customer across 5 segments via a next-best-action engine on a production batch-predict endpoint.
- Cut environment setup from hours to under 3 minutes by containerizing the full pipeline (API, dashboard, model registry) with Docker Compose.

EDUCATION

Bachelor of Artificial Intelligence — DHA Suffa University Oct 2023 – Expected 2027

CERTIFICATIONS

Microsoft Azure AI Fundamentals | Microsoft Azure Data Fundamentals | CS50x Puzzles (2025) | Speed Debugging – IEEE PSYWSC 2025 | MIT Tournament of Codes | Competitive coding: FAST (Procom '24, '26), IBA (Enigma), DSU, PAF-KIET

EXTRACURRICULAR

- Runner-up & Captain — 15th DSU Inter-Department Box Cricket Championship 2025

  Badminton · Table tennis · Football · Gym